NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Average Payoff, Typical Ticket

AP Statistics · Unit 2 · Topic 2.8 · B: 2026-10-30 · E: 2026-11-20

[STAR] TODAY'S PROBLEM

Suppose a club sells 200 raffle tickets for \$2 each. One ticket wins \$100, four tickets win \$25 each, 20 tickets win \$5 each, and the other tickets win nothing. Let X be the buyer's net profit from one randomly chosen ticket: prize minus ticket cost.

Priya: *The average gross payout is \$1.50 per ticket, so buying one is a good deal.*

Mateo: *Most tickets win nothing.*

Ticket net-profit distribution

Prize	Net x	$P(X = x)$
Top prize	98	1/200
Second prize	23	4/200
Small prize	3	20/200
No prize	-2	175/200

FIRST TAKE — before the video (5 min)

Would you buy one ticket if your goal were to make money? Use one number or outcome from the problem.

[BOOK] DISTRIBUTIONS AND ONE OUTCOME (keep on the page)

A discrete random variable assigns a number to each random outcome. Its distribution lists every possible value with a probability between 0 and 1; all probabilities add to 1. Add the probabilities of the outcomes in an event. Net profit is prize minus ticket cost. An average can differ from the most common outcome; large prizes can raise it, and it does not promise one buyer's result.

Finish after the video:

DOK 1 **(a)** State the four possible net profits X. Which is most likely?

DOK 2 **(b)** Check that the listed probabilities add to 1. Find the chance that a ticket loses money.

DOK 3 **(c)** Can Priya's average payout and Mateo's statement both be true? Use the distribution to explain. Decide whether Priya's good-deal claim follows for a buyer trying to make money, using the ticket cost and possible net profits. Write 3–4 sentences.

Frame: Both statements can be true because the few ____________ raise the average while most tickets ____________.

Frame: For a buyer trying to make money, I would ________ because the ticket costs ______ and the distribution shows ____________.

EXIT — turn this in

One thing the video changed about my first take: ______________________________